Faculté des Sciences et Technologies

(FST)

Rapport du travail de Laboratoire N° 3_Reseaux II

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Niveau : L3

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L’objectif de ce TD est de :

1. Apprendre à créer et gérer plusieurs VLANs pour isoler les différents départements (Administration, Commercial, Technique,

Invités, DMZ).

2. Comprendre le routage inter-VLAN à l’aide de routeurs et de sous-interfaces.

3. Identifier les avantages de la segmentation pour la sécurité.

4. Configurer des mots de passe forts pour l’accès console et VTY.

5. Activer SSH sur les routeurs pour un accès sécurisé à distance.

6. Appliquer la port-security sur les switchs pour limiter les adresses MAC autorisées sur chaque port.

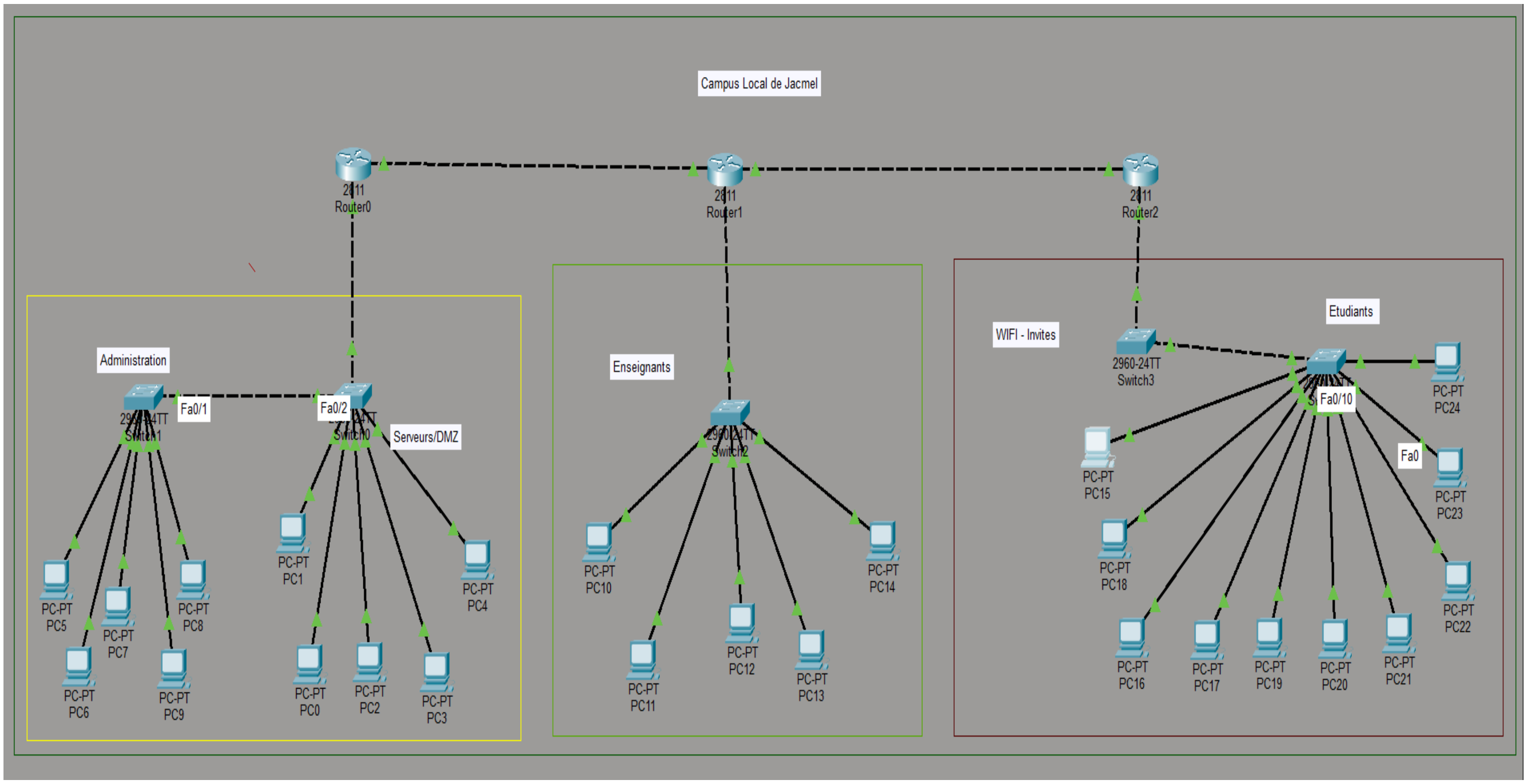
7. Créer des ACL étendues pour contrôler l’accès entre VLANs (ex. : VLAN invités ne peuvent pas accéder à la DMZ ou aux ressources sensibles).

8. Filtrer le trafic réseau de façon sélective (protocole, IP source/destination).

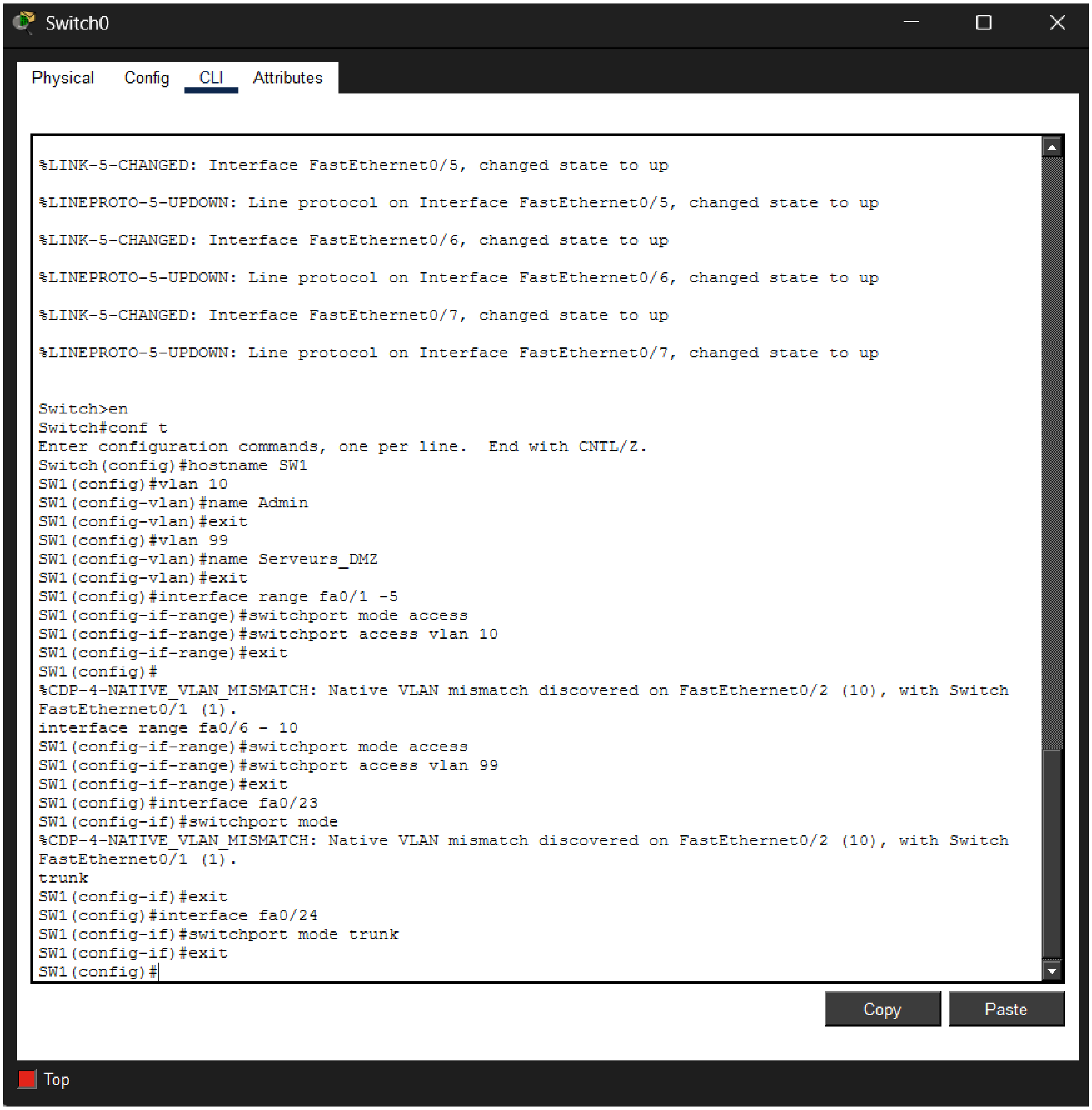
9. Simulation de scénarios d’attaque et test

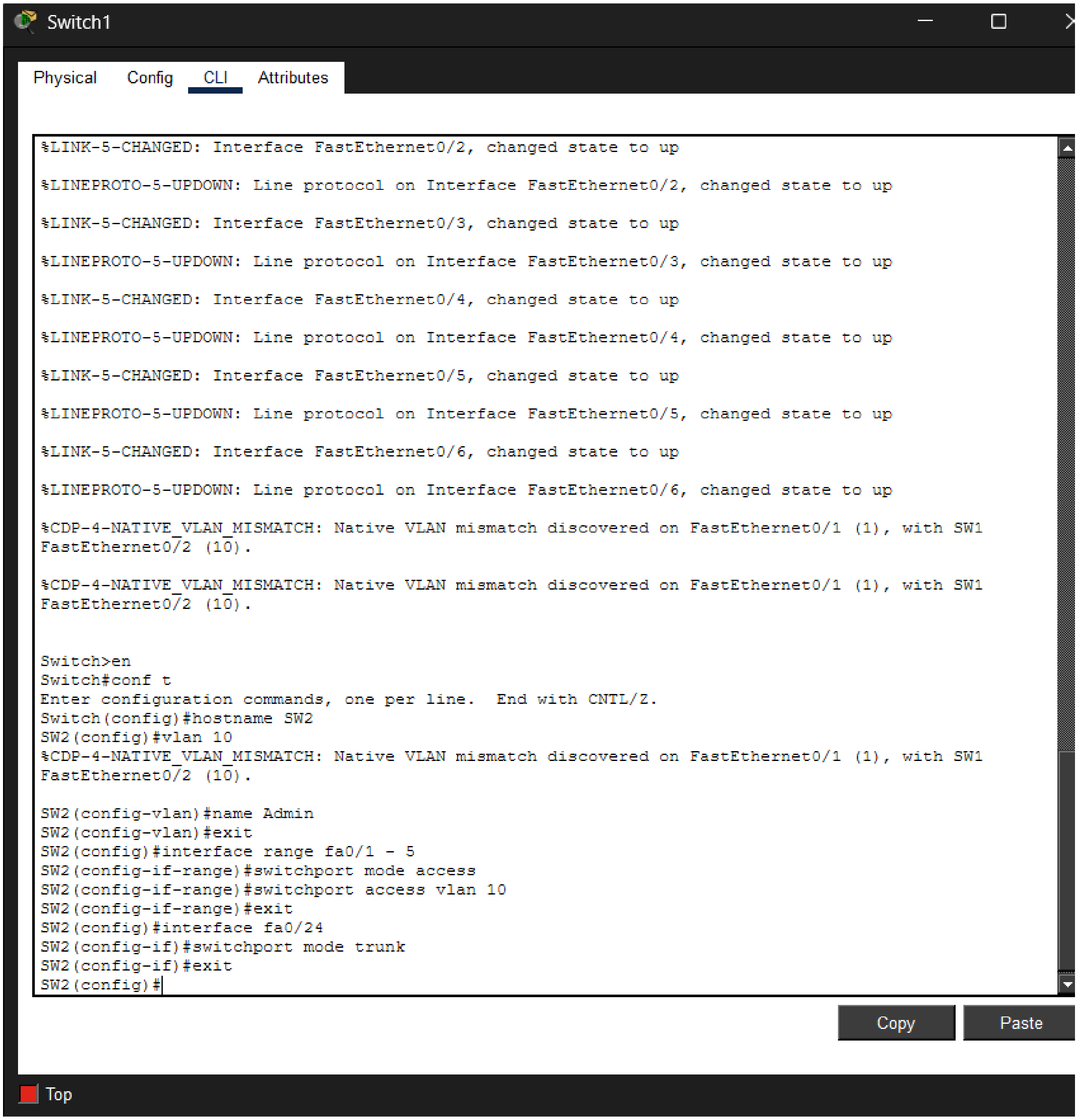
1. Création de la topologie réseau

Placer 3 routeurs, 5 switches et 25 PC sur Packet Tracer.



2. Configurer les VLANs sur les switch :




Switch1
Physical
Config
CLI
Attributes
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW1 FastEthernet0/2 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW1 FastEthernet0/2 (10).
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#vlan 10
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW1 FastEthernet0/2 (10).
SW2(config-vlan)#name Admin
SW2(config-vlan)#exit
SW2(config)#interface range fa0/1 - 5
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 10
SW2(config-if-range)#exit
SW2(config)#interface fa0/24
SW2(config-if)#switchport mode trunk
SW2(config-if)#exit
SW2(config)#
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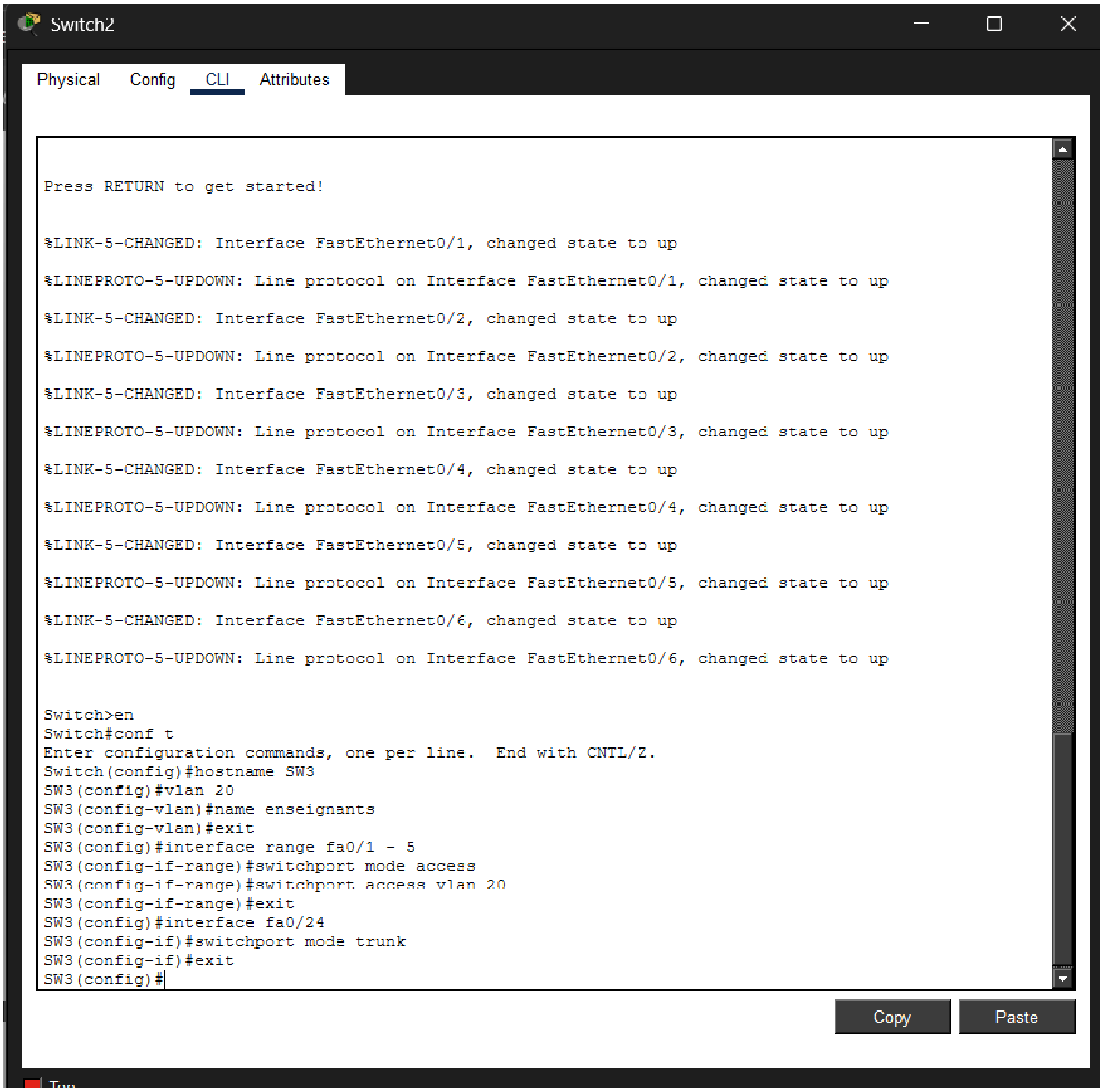
Top


Switch2
Physical
Config
CLI
Attributes
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#vlan 20
SW3(config-vlan)#name enseignants
SW3(config-vlan)#exit
SW3(config)#interface range fa0/1 - 5
SW3(config-if-range)#switchport mode access
SW3(config-if-range)#switchport access vlan 20
SW3(config-if-range)#exit
SW3(config)#interface fa0/24
SW3(config-if)#switchport mode trunk
SW3(config-if)#exit
SW3(config)#
Copy
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Switch3

Physical Config CLI Attributes

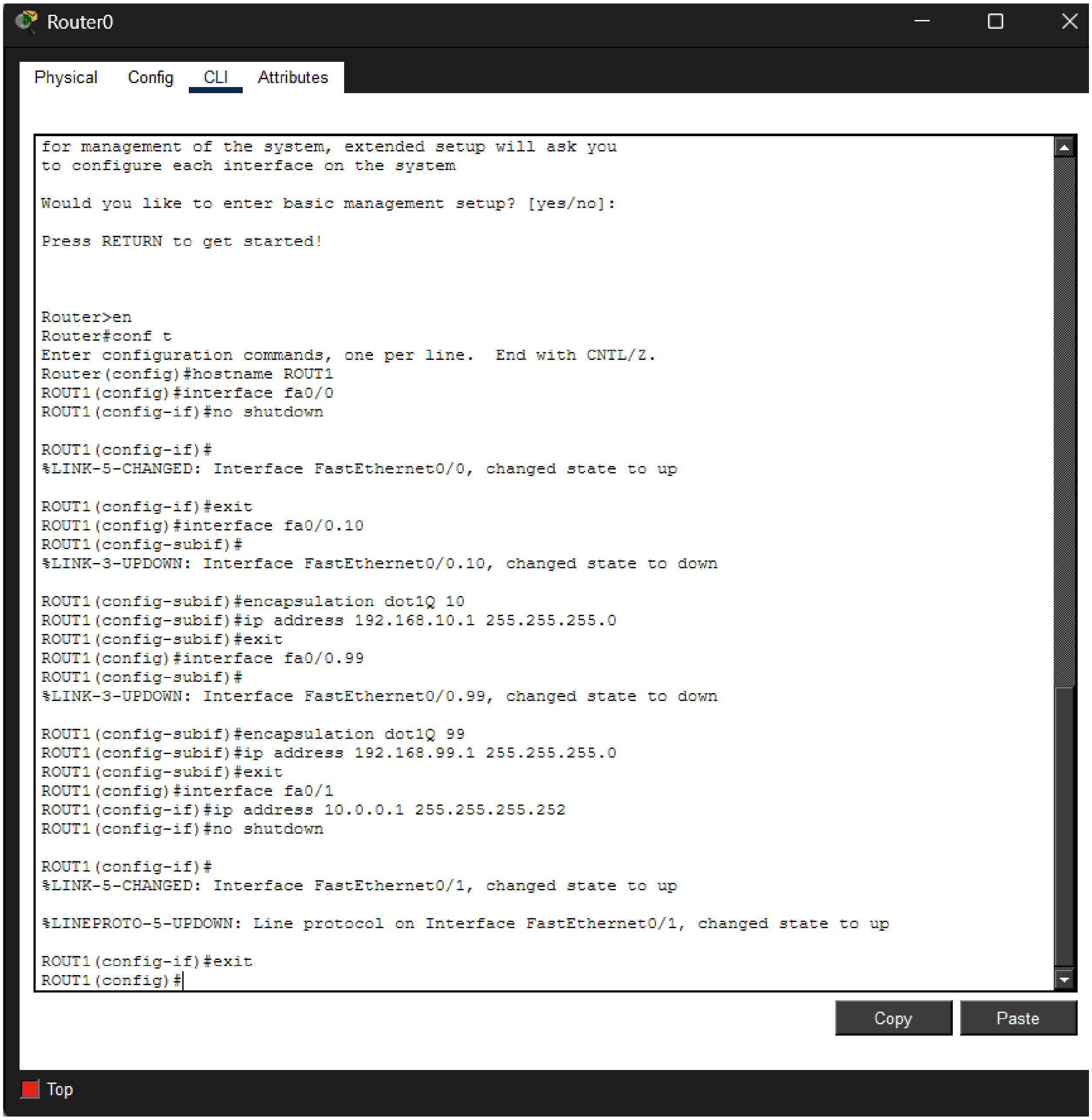
```
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#vlan 30
SW4(config-vlan)#exit
SW4(config)#vlan 40
SW4(config-vlan)#name WIFI_Invites
SW4(config-vlan)#exit
SW4(config)#interface range fa0/1 - 5
SW4(config-if-range)#switchport mode access
SW4(config-if-range)#switchport access vlan 40
SW4(config-if-range)#exit
SW4(config)#interface fa0/23
SW4(config-if)#switchport
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with Switch
FastEthernet0/1 (1).
 mode trunk
SW4(config-if)#exit
SW4(config)#interface fa0/24
SW4(config-if)#switchport mode trunk
SW4(config-if)#exit
SW4(config)#
```

Switch4

Physical Config CLI Attributes

```
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4
FastEthernet0/2 (40).


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4
FastEthernet0/2 (40).
hostname SW5
SW5(config)#vlan 30
SW5(config-vlan)#name Etudiants
SW5(config-vlan)#exit
SW5(config)#interface range fa0/1 - 5
SW5(config-if-range)#switchport mode access
SW5(config-if-range)#switch
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4
FastEthernet0/2 (40).
port access vlan 30
SW5(config-if-range)#exit
SW5(config)#interface fa0/24
SW5(config-if)#switchport mode trunk
SW5(config-if)#exit
SW5(config)#
```

3. Configurer le routage inter-VLAN



Router1

Physical Config CLI Attributes

```
Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]:

Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostmane ROUT2
                   ^
% Invalid input detected at '^' marker.

Router(config)#hostname ROUT2
ROUT2(config)#interface fa0/0
ROUT2(config-if)#no shutdown

ROUT2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

ROUT2(config-if)#exit
ROUT2(config)#interface fa0/0.20
ROUT2(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

ROUT2(config-subif)#encapsulation dot1Q 20
ROUT2(config-subif)#ip address 192.168.20.1 255.255.255.0
ROUT2(config-subif)#exit
ROUT2(config)#interface fa0/1
ROUT2(config-if)#ip address 10.0.0.2 255.255.255.252
ROUT2(config-if)#no shutdown

ROUT2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

ROUT2(config-if)#exit
ROUT2(config)#
```

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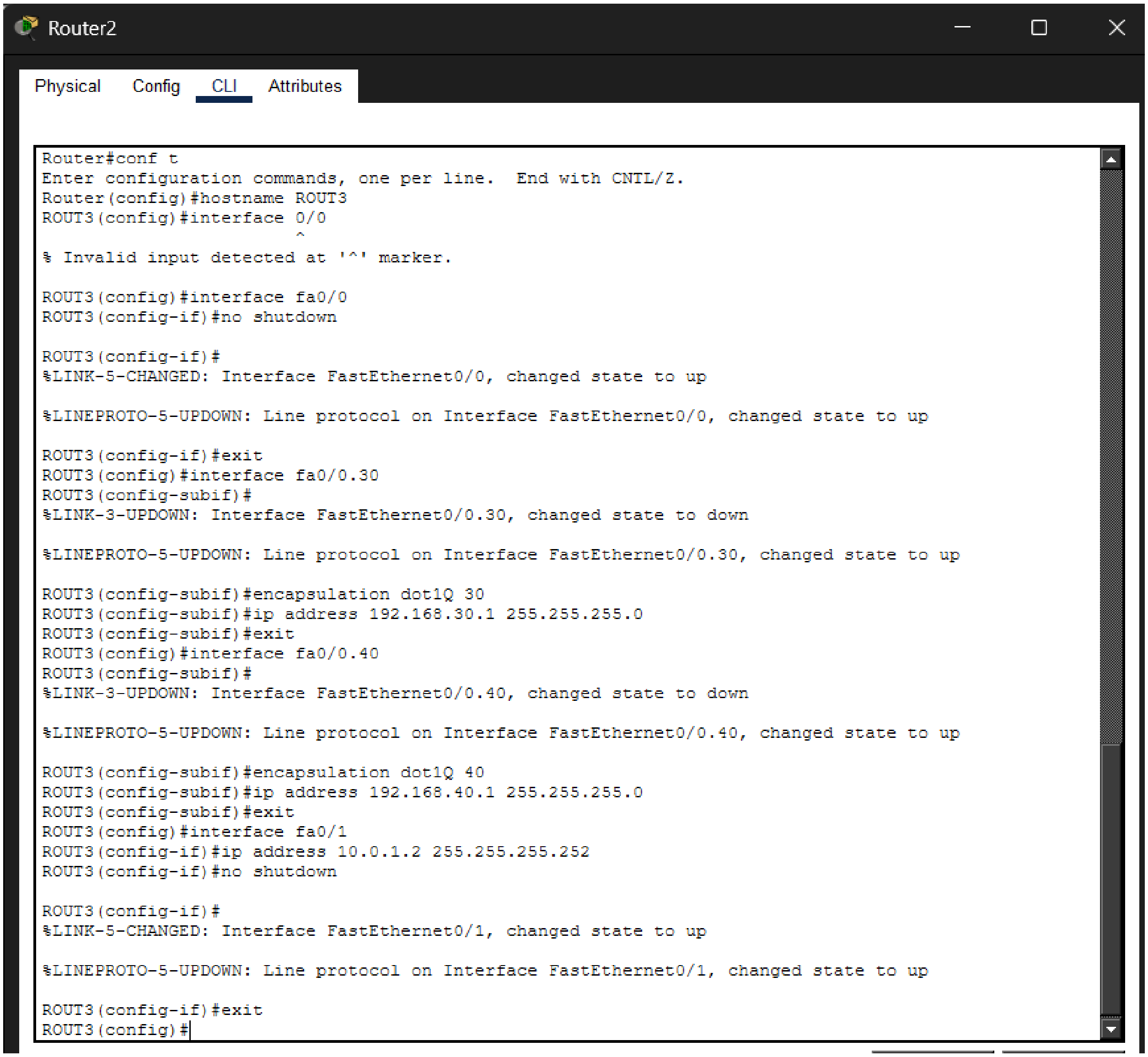
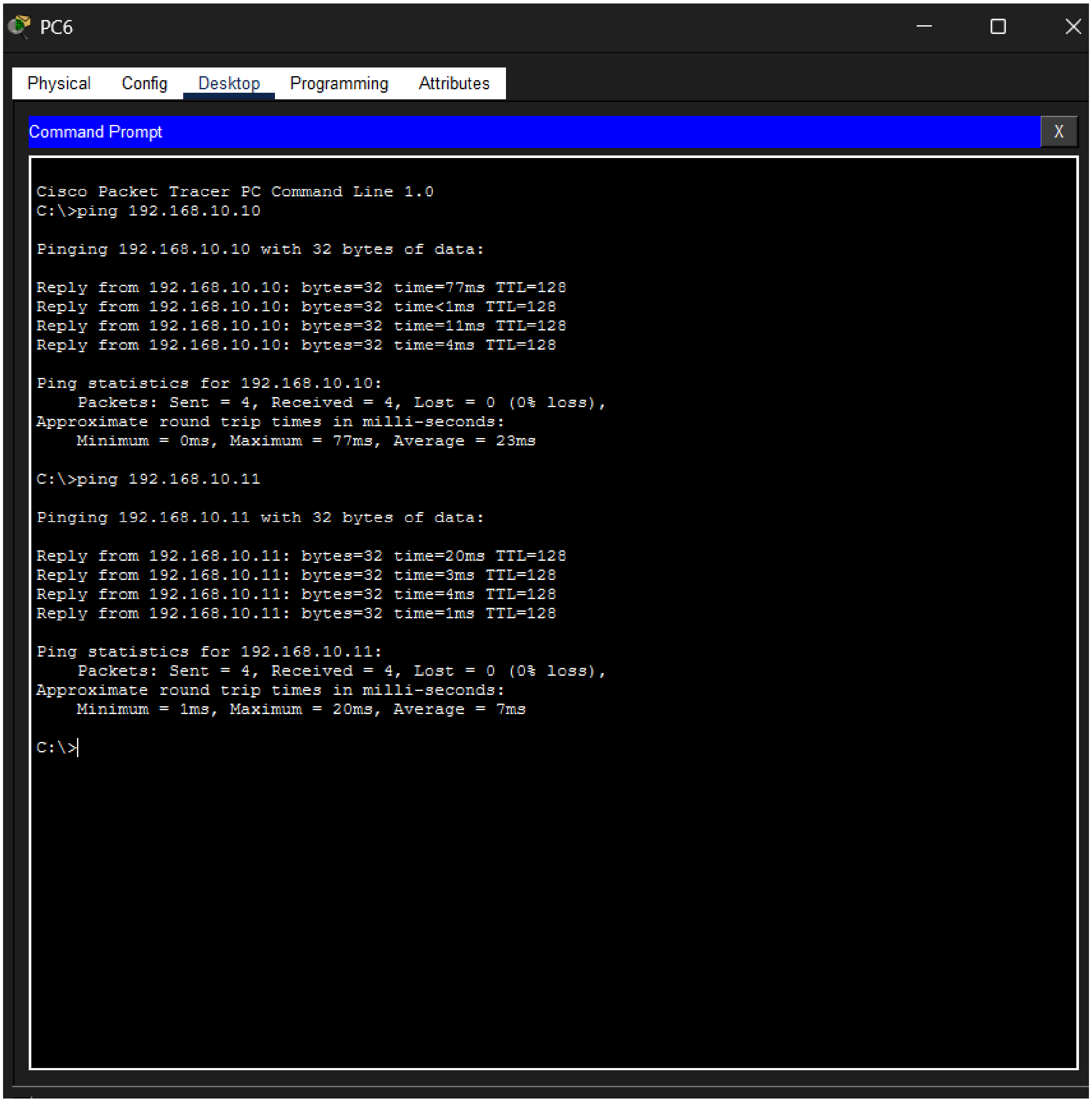
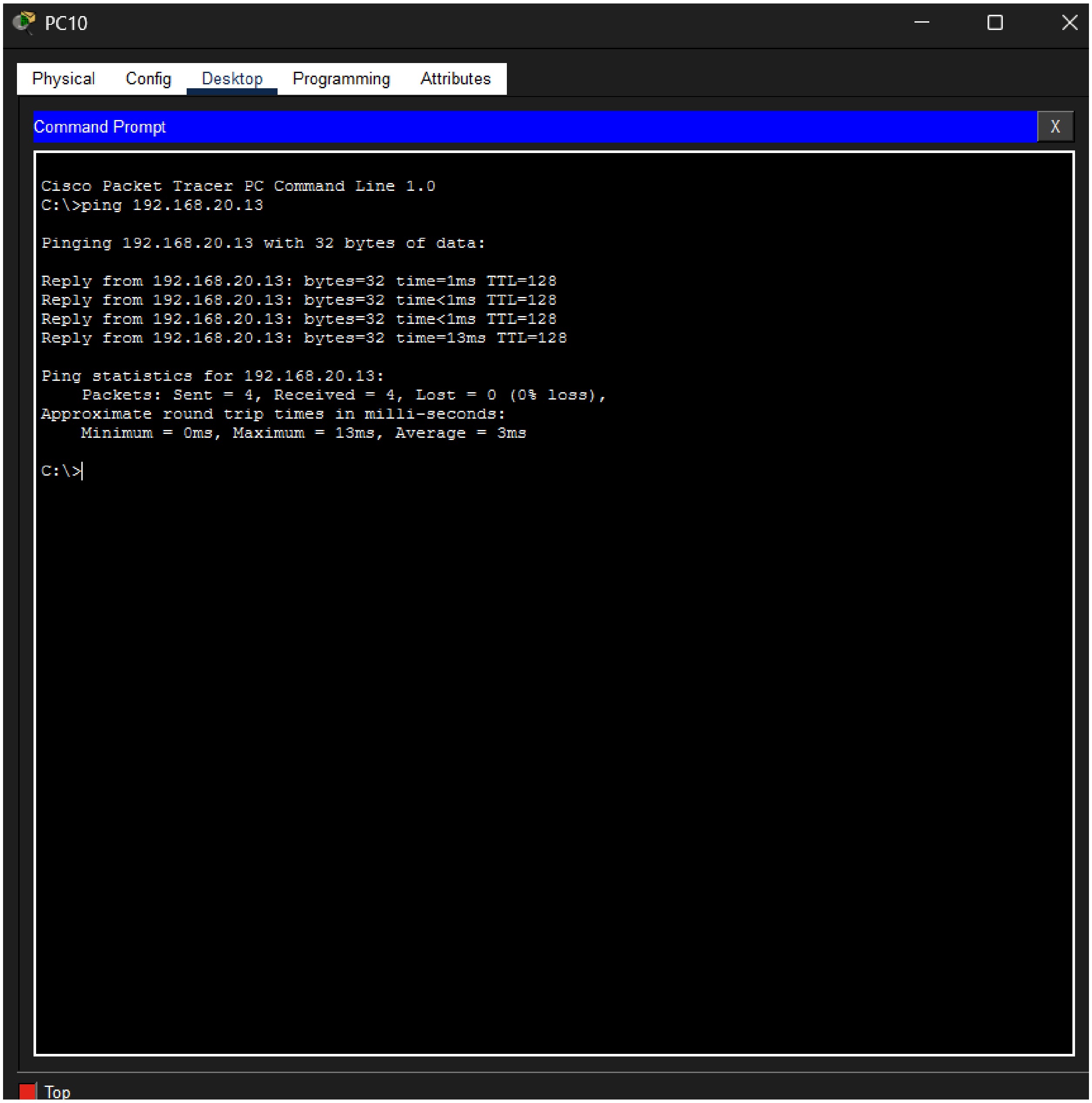

Router2
Physical
Config
CLI
Attributes
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ROUT3
ROUT3(config)#interface 0/0
^
% Invalid input detected at '^' marker.
ROUT3(config)#interface fa0/0
ROUT3(config-if)#no shutdown
ROUT3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ROUT3(config-if)#exit
ROUT3(config)#interface fa0/0.30
ROUT3(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up
ROUT3(config-subif)#encapsulation dot1Q 30
ROUT3(config-subif)#ip address 192.168.30.1 255.255.255.0
ROUT3(config-subif)#exit
ROUT3(config)#interface fa0/0.40
ROUT3(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.40, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.40, changed state to up
ROUT3(config-subif)#encapsulation dot1Q 40
ROUT3(config-subif)#ip address 192.168.40.1 255.255.255.0
ROUT3(config-subif)#exit
ROUT3(config)#interface fa0/1
ROUT3(config-if)#ip address 10.0.1.2 255.255.255.252
ROUT3(config-if)#no shutdown
ROUT3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ROUT3(config-if)#exit
ROUT3(config)#

- **Vérification de la connectivité avec ping.**




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=77ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=11ms TTL=128
Reply from 192.168.10.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 77ms, Average = 23ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=20ms TTL=128
Reply from 192.168.10.11: bytes=32 time=3ms TTL=128
Reply from 192.168.10.11: bytes=32 time=4ms TTL=128
Reply from 192.168.10.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 7ms

C:\>
```



4. Mis en place de la sécurité réseau

1. ACL (Access Control List)

- Bloquer certains VLAN (ex. invités) d'accéder à la DMZ ou à l'administration.

```
ROUT3>en
ROUT3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ROUT3(config)#access-list 120 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
ROUT3(config)#access-list 120 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
ROUT3(config)#access-list 120 permit ip any any
ROUT3(config)#interface fa0/0.40
ROUT3(config-subif)#ip access-group 120 in
ROUT3(config-subif)#exit
ROUT3(config)#
```

```
ROUT1>en
ROUT1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ROUT1(config)#access-list 110 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
ROUT1(config)#access-list 110 permit ip any any
ROUT1(config)#interface fa0/0.99
ROUT1(config-subif)#ip access-group 110 in
ROUT1(config-subif)#exit
ROUT1(config)#
```

2. Port Security

- Limiter le nombre d'appareils par port sur les switches.
- Configurer la violation pour restreindre l'accès si un PC non autorisé se connecte.

```
SW1(config)#interface range fa0/1 - 22
SW1(config-if-range)#switchport port-security maximum 2
SW1(config-if-range)#switchport port-security
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
SW1(config-if-range)#switchport port-security violation restrict
SW1(config-if-range)#switchport port-security mac-address sticky
SW1(config-if-range)#exit
SW1(config)#
```

```
SW2>en
SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#interface range fa0/1 - 22
SW2(config-if-range)#switchport port-security maximum 2
SW2(config-if-range)#switchport port-security violation restrict
SW2(config-if-range)#switchport port-security mac-address sticky
SW2(config-if-range)#exit
SW2(config)#
```

```
SW3>en
SW3# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#interface range fa0/1 - 22
SW3(config-if-range)#switchport port-security maximum 2
SW3(config-if-range)#switchport port-security violation restrict
SW3(config-if-range)#switchport port-security mac-address sticky
SW3(config-if-range)#exit
SW3(config)#
```

```
SW4>en
SW4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#interface range fa0/1 - 22
SW4(config-if-range)#switchport port-security maximum 2
SW4(config-if-range)#switchport port-security violation restrict
SW4(config-if-range)#switchport port-security mac
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with SW5
FastEthernet0/1 (30).
-address sticky
SW4(config-if-range)#exit
SW4(config)#
```

```
SW5>en
SW5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW5(config)#interface range fa0/1 - 22
SW5(config-if-range)#switchport port-security maximum 2
SW5(config-if-range)#switchport port-security viols
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (30), with SW4
FastEthernet0/2 (40).ation restric
SW5(config-if-range)#switchport port-security mac-address sticky
SW5(config-if-range)#exit
SW5(config)#
```

3. Sécurisation des routeurs

- Désactiver Telnet, activer SSH pour l'administration.
- Configurer des mots de passe forts.

```
ROUT1>en
ROUT1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ROUT1(config)#ip domain-name campus.local
ROUT1(config)#crypto key generate rsa
The name for the keys will be: ROUT1.campus.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ROUT1(config)#username admin privilege 15 secret Cisco@123
*Mar 2 1:19:51.147: %SSH-5-ENABLED: SSH 1.99 has been enabled
ROUT1(config)#line vty 0 4
ROUT1(config-line)#transport input ssh
ROUT1(config-line)#login local
ROUT1(config-line)#exit
ROUT1(config)#lone console 0
                ^
% Invalid input detected at '^' marker.

ROUT1(config)#line console 0
ROUT1(config-line)#password Cisco@123
ROUT1(config-line)#login
ROUT1(config-line)#exit
ROUT1(config)#line vty 0 4
ROUT1(config-line)#transport input shh
                                    ^
% Invalid input detected at '^' marker.

ROUT1(config-line)#transport input ssh
ROUT1(config-line)#exit
ROUT1(config)#
```

```
ROUT2>en
ROUT2# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ROUT2(config)#ip domain-name campus.local
ROUT2(config)#crypto key generate rsa
The name for the keys will be: ROUT2.campus.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ROUT2(config)#username admin privilege 15 secret Cisco@123
*Mar 2 1:23:17.855: %SSH-5-ENABLED: SSH 1.99 has been enabled
ROUT2(config)#line vty 0 4
ROUT2(config-line)#transport input ssh
ROUT2(config-line)#login local
ROUT2(config-line)#exit
ROUT2(config)#line console 0
ROUT2(config-line)#password Cisco@123
ROUT2(config-line)#login
ROUT2(config-line)#exit
ROUT2(config)#line vty 0 4
ROUT2(config-line)#transport input ssh
ROUT2(config-line)#exit
ROUT2(config)#
```

```
ROUT3>en
ROUT3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ROUT3(config)#ip domain-name campus.local
ROUT3(config)#crypto key generate rsa
The name for the keys will be: ROUT3.campus.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ROUT3(config)#username admin privilege 15 secret Cisco@123
*Mar 2 1:26:7.569: %SSH-5-ENABLED: SSH 1.99 has been enabled
ROUT3(config)#line vty 0 4
ROUT3(config-line)#transport input ssh
ROUT3(config-line)#login local
ROUT3(config-line)#exit
ROUT3(config)#lone console 0
                ^
% Invalid input detected at '^' marker.

ROUT3(config)#line console 0
ROUT3(config-line)#password Cisco@123
ROUT3(config-line)#login
ROUT3(config-line)#exit
ROUT3(config)#line vty 0 4
ROUT3(config-line)#transport input ssh
ROUT3(config-line)#xit
                   ^
% Invalid input detected at '^' marker.

ROUT3(config-line)#exit
ROUT3(config)#
```

Scénarios de test

- Ping entre VLAN autorisés : admin → enseignants → doit fonctionner.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.13

Pinging 192.168.20.13 with 32 bytes of data:

Reply from 192.168.20.13: bytes=32 time=1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.20.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

- Ping VLAN invités → serveur DMZ : doit échouer.

PC17

Physical Config Desktop Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.99.12

Pinging 192.168.99.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.99.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- tester que Telnet est désactivé

```
PC10
Physical   Config   Desktop   Programming   Attributes
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.13

Pinging 192.168.20.13 with 32 bytes of data:

Reply from 192.168.20.13: bytes=32 time=1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time<1ms TTL=128
Reply from 192.168.20.13: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.20.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>telnet 192.168.20.10
Trying 192.168.20.10 ...
% Connection refused by remote host
C:\>
```

```
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.10.13
Trying 192.168.10.13 ...
% Connection timed out; remote host not responding
C:\>
```

- Connexion avec ssh

```
ROUT2#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
ROUT2#
```

- Vérification sur le swicth

```
SW1>en
SW1#show port-security interface fa0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0

SW1#
```

Conclusion

Ce TD m'a permis de comprendre et de mettre en pratique la segmentation réseau par VLANs, le routage inter-VLAN via des sous-interfaces, ainsi que la sécurisation des accès par SSH et Port Security. La simulation sur Cisco Packet Tracer a été un outil essentiel pour valider chaque configuration et tester les scénarios de sécurité.